

U.S.N.

B.M.S. College of Engineering, Bengaluru-560019

Autonomous Institute Affiliated to VTU

August / September 2025 Supplementary Examinations

Programme: B.E.

Semester: VI

Branch: Computer Science and Engineering

Duration: 3 hrs.

Course Code: 23CS6PCMAL

Max Marks: 100

Course: Machine Learning

Instructions: 1. Answer any FIVE full questions, choosing one full question from each unit.
2. Missing data, if any, may be suitably assumed.

Important Note: Completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages. Revealing of identification, appeal to evaluator will be treated as malpractice.			UNIT - I	CO	PO	Marks
	1	a)	List and explain the main challenges of Machine Learning (ML).	CO1	PO1	6
		b)	Analyse the following statements given below. For each of the statement, identify the type of machine learning required to be applied. Justify your answer. i. Identify patterns and relationships in the data ii. Rule-based ML technique that finds out some very useful relations between parameters of a large data set. iii. The model is trained with a combination of labeled and unlabeled datasets	CO2	PO2	6
		c)	Consider a scenario where you are working as a Machine Learning Engineer at an insurance company. Your team is developing a fraud detection system to identify whether an insurance claim is fraudulent or not. The dataset contains historical claim data with features such as claim amount, customer age, policy type, number of previous claims, and claim approval time. Answer the following questions with suitable answer. • What is the nature of the problem? • What are the key characteristics of the dataset? • What visualization techniques would you use? • How would you handle missing values?	CO3	PO3	8
			OR			
	2	a)	With diagram bring out the difference between Traditional programming approach and machine learning approach.	CO1	PO1	6
		b)	Discuss the types of machine learning with examples.	CO1	PO1	8
		c)	Explain Minmax scalar and Standard Scalar with an example.	CO2	PO2	6

		UNIT - II																																				
3	a)	Consider a binary classification problem where we want to predict whether a student will pass or fail based on their study hours. The logistic regression model has been trained, and the learned parameters are $a_0 = -6$ (intercept) and $a_1 = 0.9$ (coefficient for study hours). i) Write the logistic regression equation for this problem. ii) Calculate the probability that a student who studies for 7 hours will pass. iii) Determine the predicted class (pass or fail) for this student based on a threshold of 0.6.	CO3	PO2	10																																	
	b)	Apply the K-Nearest Neighbor (KNN) to predict the diabetic patient with the given feature BMI and Age. Assume K=3, predict the value of Sugar for BMI=43.6, Age=40 <table border="1"><thead><tr><th>BMI</th><th>Age</th><th>Sugar</th></tr></thead><tbody><tr><td>33.6</td><td>50</td><td>1</td></tr><tr><td>26.6</td><td>30</td><td>0</td></tr><tr><td>23.4</td><td>40</td><td>0</td></tr><tr><td>43.1</td><td>67</td><td>0</td></tr><tr><td>35.3</td><td>23</td><td>1</td></tr><tr><td>35.9</td><td>67</td><td>1</td></tr><tr><td>36.7</td><td>45</td><td>1</td></tr><tr><td>25.7</td><td>46</td><td>9</td></tr><tr><td>23.3</td><td>29</td><td>0</td></tr><tr><td>31</td><td>56</td><td>1</td></tr></tbody></table>	BMI	Age	Sugar	33.6	50	1	26.6	30	0	23.4	40	0	43.1	67	0	35.3	23	1	35.9	67	1	36.7	45	1	25.7	46	9	23.3	29	0	31	56	1	CO3	PO2	10
BMI	Age	Sugar																																				
33.6	50	1																																				
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25.7	46	9																																				
23.3	29	0																																				
31	56	1																																				
		OR																																				
4	a)	Define the following i) Entropy ii) Information Gain	CO1	PO1	4																																	
	b)	Given with eight data points, out of which four are positively labelled and four are negatively labelled. Support Vector-1 (Positive): {(3,1), (3,-1), (6,1), (6, -1)} Support Vector-2 (Negative): {(1,0), (0,1), (0,-1), (-1, 0)} i. Using the given support vectors, calculate the Weight vectors and bias ii. Find the Hyperplane	CO1	PO1	8																																	
	c)	Apply Linear Regression for the values given in the following table. The table shows week and sales data. Use matrix approach. Predict the value of y for x=5.	CO3	PO3	8																																	

		<table><tr><th>X_i(week)</th><th>Y_j(Sales in thousands)</th></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>3</td></tr><tr><td>3</td><td>4</td></tr><tr><td>4</td><td>8</td></tr></table>	X _i (week)	Y _j (Sales in thousands)	1	1	2	3	3	4	4	8												
X _i (week)	Y _j (Sales in thousands)																							
1	1																							
2	3																							
3	4																							
4	8																							
		UNIT - III																						
5	a)	Differentiate between ROC (Receiver Operating Characteristic) curves and AUC (Area Under the Curve).	CO1	PO1	6																			
	b)	Explain different types of cross validation methods.	CO2	PO2	6																			
	c)	Consider the following table <table><tr><th>Test Split</th><th>Model A Accuracy(%)</th><th>Model B Accuracy (%)</th></tr><tr><td>1</td><td>85</td><td>83</td></tr><tr><td>2</td><td>87</td><td>85</td></tr><tr><td>3</td><td>90</td><td>88</td></tr><tr><td>4</td><td>86</td><td>87</td></tr><tr><td>5</td><td>88</td><td>86</td></tr></table> Perform paired t- test on the above table. With 95% confidence level and the critical t-value for a two-tailed test is 2.776. Predict the model hypothesis.	Test Split	Model A Accuracy(%)	Model B Accuracy (%)	1	85	83	2	87	85	3	90	88	4	86	87	5	88	86	CO3	PO3	8	
Test Split	Model A Accuracy(%)	Model B Accuracy (%)																						
1	85	83																						
2	87	85																						
3	90	88																						
4	86	87																						
5	88	86																						
		OR																						
6	a)	List and explain the different guidelines for machine learning experiments.	CO1	PO1	10																			
	b)	Consider the following Confusion Matrix of Multiclass classification with class labels as (1,2,3). Calculate Accuracy, Precision, Recall and F1-Score for each class. <table><tr><th rowspan="2">True Class</th><th colspan="3">Predicted Class</th></tr><tr><th>1</th><th>2</th><th>3</th></tr><tr><td>1</td><td>8</td><td>2</td><td>0</td></tr><tr><td>2</td><td>1</td><td>9</td><td>0</td></tr><tr><td>3</td><td>1</td><td>2</td><td>7</td></tr></table>	True Class	Predicted Class			1	2	3	1	8	2	0	2	1	9	0	3	1	2	7	CO3	PO3	10
True Class	Predicted Class																							
	1	2	3																					
1	8	2	0																					
2	1	9	0																					
3	1	2	7																					
		UNIT - IV																						
7	a)	Explain Boosting, Bagging and Stacking with example.	CO1	PO1	10																			
	b)	Cluster the following eight points (with (x, y) representing locations) into three clusters: A1(2, 10), A2(2, 5), A3(8, 4), A4(5, 8), A5(7, 5), A6(6, 4), A7(1, 2), A8(4, 9)	CO3	PO3	10																			

			Initial cluster centers are: A1(2, 10), A4(5, 8) and A7(1, 2). The distance function between two points a = (x1, y1) and b = (x2, y2) is defined as P(a, b) = x2 – x1 + y2 – y1 Use K-Means Clustering algorithm to find the three cluster centers after the second iteration.																		
			OR																		
	8	a)	Describe two different methods of voting in Ensemble learning with an example.	CO1	PO1	10															
		b)	Define Dimensionality reduction. Given the data in table, reduce the dimension from two to one using Principal Component Analysis (PCA) <table border="1"><tr><td>Feature</td><td>Example1</td><td>Example2</td><td>Example3</td><td>Example 4</td></tr><tr><td>X1</td><td>4</td><td>8</td><td>13</td><td>7</td></tr><tr><td>X2</td><td>11</td><td>4</td><td>5</td><td>14</td></tr></table>	Feature	Example1	Example2	Example3	Example 4	X1	4	8	13	7	X2	11	4	5	14	CO3	PO3	10
Feature	Example1	Example2	Example3	Example 4																	
X1	4	8	13	7																	
X2	11	4	5	14																	
			UNIT - V																		
	9	a)	Define Markov decision process (MDP). Explain the components of MDP with an example.	CO1	PO1	8															
		b)	Explain Reinforcement learning with diagram and learning task.	CO1	PO1	6															
		c)	Explain Generalization in Reinforcement learning.	CO1	PO1	6															
			OR																		
	10	a)	Explain Model Based and Model free Reinforcement Learning Algorithms.	CO1	PO1	10															
		b)	i. Explain Q- learning Algorithm ii. Consider the scenario: - Agent is in state S3, Takes action a2, Lands in state S4 - Gets reward R=5, Learning rate $\alpha=0.6$, Discount factor $\gamma=0.9$ Solve for (S3, a2). Show all the steps. <table border="1"><tr><td>State - action</td><td>a1</td><td>a2</td><td>a3</td><td>a4</td></tr><tr><td>S3</td><td>3</td><td>6</td><td>1</td><td>2</td></tr><tr><td>S4</td><td>0</td><td>4</td><td>5</td><td>8</td></tr></table>	State - action	a1	a2	a3	a4	S3	3	6	1	2	S4	0	4	5	8	CO3	PO3	10
State - action	a1	a2	a3	a4																	
S3	3	6	1	2																	
S4	0	4	5	8																	
